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Linearly dependent & linearly independent vectors:- so

The vectors $v_1, v_2, \dots, v_m$ of a vector space V over F are said to be linearly dependent if there exist elements $a_1, a_2, \dots, a_m \in F$, not all zeros such that

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0$$

On the other hand if

$$a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0 \quad \text{where all } a_i = 0$$

then vectors $v_1, v_2, \dots, v_m$ are said to be linearly independent.

Note

① If any two vectors out of $v_1, v_2, \dots, v_m$ are equal say $v_3 = v_4$

then $v_1, v_2, \dots, v_m$ are linearly dependent

$$\text{because } 0 \cdot v_1 + 0 \cdot v_2 + 1 \cdot v_3 + (-1) \cdot v_4 + 0 \cdot v_5 + \dots + 0 \cdot v_m = 0$$

② If any one of the vector out of $v_1, v_2, \dots, v_m$ is zero, say $v_2 = 0$ then $v_1, v_2, \dots, v_m$ are linearly dependent because

$$0 \cdot v_1 + 1 \cdot v_2 + 0 \cdot v_3 + \dots + 0 \cdot v_m = 0$$

③ A single non zero vector is always linearly independent because

let $v \neq 0$ be the single vector

$$\text{then } a v = 0 \Rightarrow a = 0$$

Thus v is linearly independent.

④ Two vectors v_1 & v_2 are linearly dependent if one of them is a multiple of other.

Theorem Let V be a vector space over a field F .
 $\& S = \{v_1, v_2, \dots, v_m\}$ be a set of vectors in V .
 Then

- (i) If S is linearly independent then any subset of S is also linearly independent.
 (ii) If S is linearly dependent, then the set $\{v, v_1, v_2, \dots, v_m\}$ is linearly dependent for all $v \in V$.
 i.e., every superset of S is also linearly dependent.

Proof:

(i) Here $S = \{v_1, v_2, \dots, v_m\}$

Let $\{v_1, v_2, \dots, v_i\}$ where $i < m$
 is a subset of S

$\&$ let $a_1 v_1 + a_2 v_2 + \dots + a_i v_i = 0$ where $a_i \in F$

or $a_1 v_1 + a_2 v_2 + \dots + a_i v_i + 0 v_{i+1} + \dots + 0 v_m = 0$

But $S = \{v_1, v_2, \dots, v_m\}$ is linearly dependent

So $a_1 = a_2 = a_3 = \dots = a_i = 0$

Hence $\{v_1, v_2, \dots, v_i\}$ is linearly dependent.

(ii) As $S = \{v_1, v_2, \dots, v_m\}$ is linearly dependent

So $a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0$ where $a_i \neq 0$ for some i

Now

$0 \cdot v + a_1 v_1 + a_2 v_2 + \dots + a_m v_m = 0$ where $a_i \neq 0$ for some i

So $\{v, v_1, v_2, \dots, v_m\}$ is linearly dependent.

Theorem A set $S = \{v_1, v_2, \dots, v_n\}$ of n vectors ($n \geq 2$) in a vector space V is linearly dependent iff. at least one of the vectors in S is a linear combination of the remaining vectors of the set.

Proof:

(i) Suppose the set

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$S = \{v_1, v_2, \dots, v_n\}$ is linearly independent

Then there exist scalars $a_1, a_2, \dots, a_n$ at least one of them say a_i is non zero, such that

$$a_1 v_1 + a_2 v_2 + \dots + a_i v_i + \dots + a_n v_n = 0$$

$$\text{or } a_i v_i = -a_1 v_1 - a_2 v_2 - \dots - a_{i-1} v_{i-1} - a_{i+1} v_{i+1} - \dots - a_n v_n$$

or

$$v_i = -\frac{a_1}{a_i} v_1 - \frac{a_2}{a_i} v_2 - \dots - \frac{a_{i-1}}{a_i} v_{i-1} - \frac{a_{i+1}}{a_i} v_{i+1} - \dots - \frac{a_n}{a_i} v_n$$

which shows that v_i is a linear combination of remaining vectors of the set.

Conversely

Let some vector v_j of the given set is a linear combination of the remaining vectors. i.e.,

$$v_j = a_1 v_1 + a_2 v_2 + \dots + a_{j-1} v_{j-1} + a_{j+1} v_{j+1} + \dots + a_n v_n$$

Then above eq. can be written as

$$a_1 v_1 + a_2 v_2 + \dots + a_{j-1} v_{j-1} + (-1) v_j + a_{j+1} v_{j+1} + \dots + a_n v_n = 0$$

Here there is at least one coefficient namely -1 of v_j

which is non zero & so the set

$\{v_1, v_2, \dots, v_{j-1}, v_j, v_{j+1}, \dots, v_n\}$ is linearly dependent.

(ii) Suppose that the set $S = \{v_1, v_2, \dots, v_n\}$ is linearly dependent then there exist scalars

$a_1, a_2, \dots, a_n \in F$ not all zero such that

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0 \quad \text{--- (1)}$$

Let a_k be the last non zero scalar in (1) then

the terms $a_{k+1} v_{k+1}, a_{k+2} v_{k+2}, \dots, a_n v_n$ are all zero

so eq. (1) becomes

$$a_1 v_1 + a_2 v_2 + \dots + a_k v_k = 0 \quad \text{where } a_k \neq 0$$

$$\text{or } a_k v_k = -a_1 v_1 - a_2 v_2 - \dots - a_{k-1} v_{k-1}$$

$$\text{or } v_k = -\frac{a_1}{a_k} v_1 - \frac{a_2}{a_k} v_2 - \dots - \frac{a_{k-1}}{a_k} v_{k-1}$$

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Which shows that V_k is a linear combination of the vectors preceding it.

Conversely

Suppose that in $S = \{V_1, V_2, \dots, V_n\}$, some of the vectors say V_k is a linear combination of the vectors preceding it

$$\text{i.e., } V_k = b_1 V_1 + b_2 V_2 + \dots + b_{k-1} V_{k-1}$$

$$\text{or } b_1 V_1 + b_2 V_2 + \dots + b_{k-1} V_{k-1} + (-1) V_k = 0$$

$$\text{or } b_1 V_1 + b_2 V_2 + \dots + b_{k-1} V_{k-1} + (-1) V_k + 0 V_{k+1} + \dots + 0 V_n = 0$$

Here atleast one coefficient -1 of V_k is non zero

Hence $S = \{V_1, V_2, \dots, V_n\}$ is linearly dependent.

Basis of a vector space:

A linearly independent set which generates or spans a vector space V is called a basis for V .

Theorem Any finite dimensional vector space contains a basis.

Proof 1.

Let V be a finite dimensional vector space then V should be linear span of some finite set. Let $\{V_1, V_2, \dots, V_r\}$ be a finite spanning set of V . On case $V_1, V_2, \dots, V_n$ are linearly independent, then they form a basis for V & the proof is complete.

Suppose $V_1, V_2, \dots, V_r$ are not linearly independent i.e; they are linearly dependent, so one of the vectors V_i is a linear combination of the preceding vectors. We drop this vector V_i from the set & obtain a set of $r-1$ vectors

$v_1, v_2, \dots, v_{i-1}$. clearly ⁵⁵⁹ any linear combination ⁵⁴ of $v_1, v_2, \dots, v_i$ is also a linear combination of $v_1, v_2, \dots, v_{i-1}$. So $\{v_1, v_2, \dots, v_{i-1}\}$ is also a spanning set for V .

Continuing in this way, we arrive at a linearly independent spanning set $\{v_1, v_2, \dots, v_n\}; 1 \leq n \leq i$ & so it forms a basis for V .

Thus every finite dimensional vector space contains a basis.

Note If a vector space V is generated by $v_1, v_2, \dots, v_m$ then any linearly independent set in V cannot have more than m no. of elements.

Dimension of a vector space:

The no. of elements in a basis of a vector space V over F is called dimension of V . It is denoted by $\dim V$.

Theorem All bases of a finite dimensional vector space contains the same no. of elements.

Proof. Let a vector space V over F has two bases A & B with m & n no. of elements.

Since A spans V & B is a linearly independent subset in V , so B cannot have more than m no. of elements.

$$\text{i.e., } n \leq m \quad \text{---} \quad \textcircled{1}$$

Now since B spans V & A is a linearly independent subset in V , so A cannot have

more than n no. of elements

i.e., $m \leq n$ ————— (2)

from (1) & (2)

$$m = n$$

Hence no. of elements in A = no. of elements in B
which is req. proof.

Theorem Let V be a vector space such that $\dim V = n < \infty$.

A set of vectors $\{v_1, v_2, \dots, v_n\} \subset V$ is a basis for V iff. each vector in V is uniquely expressible as a linear combination of vectors $v_1, v_2, \dots, v_n$.

Proof. Let the set $\{v_1, v_2, \dots, v_n\}$ be a basis for V . Then every vector $v \in V$ can be expressed in at least one way as a linear combination of the basis vectors. i.e.,

$$v = a_1 v_1 + a_2 v_2 + \dots + a_n v_n \quad \text{where } a_i \in F \quad 1 \leq i \leq n$$

Suppose v can also be expressed as

$$v = b_1 v_1 + b_2 v_2 + \dots + b_n v_n \quad \text{where } b_i \in F \quad 1 \leq i \leq n$$

Comparing above eqs.

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = b_1 v_1 + b_2 v_2 + \dots + b_n v_n$$

$$\Rightarrow (a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \dots + (a_n - b_n)v_n = 0$$

Since $v_1, v_2, \dots, v_n$ are linearly independent

$$\text{So } a_1 - b_1 = 0, a_2 - b_2 = 0, \dots, a_n - b_n = 0$$

$$\Rightarrow a_1 = b_1, a_2 = b_2, \dots, a_n = b_n$$

Hence every vector $v \in V$ can be expressed in a unique way as a linear combination of $v_1, v_2, \dots, v_n$.

Conversely

Let every vector $v \in V$ is uniquely expressible as a linear combination of $v_1, v_2, \dots, v_n$.

Then these vectors span V . We will prove that S they are linearly independent.

Suppose that for scalars $a_1, a_2, \dots, a_n$

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0 \quad \text{--- (1)}$$

$$\text{As } 0v_1 + 0v_2 + \dots + 0v_n = 0 \quad \text{--- (2)}$$

Since the representation is unique

$$\text{So } a_1 = 0, a_2 = 0, \dots, a_n = 0$$

Hence $v_1, v_2, \dots, v_n$ are linearly independent.

Since they also span V .

Hence $\{v_1, v_2, \dots, v_n\}$ form a basis for V

Theorem Let $S = \{v_1, v_2, \dots, v_n\}$ be a basis for an n -dimensional vector space V over a field F .

Then every set with more than n vectors is linearly dependent.

Proof:

Let $B = \{u_1, u_2, \dots, u_r\}$ be a set of r vectors in V where $r > n$.

We shall show that B is linearly dependent.

To show that B is linearly dependent, we must find scalars $c_1, c_2, \dots, c_r$, not all zero, such that

$$c_1 u_1 + c_2 u_2 + \dots + c_r u_r = 0 \quad \text{--- (1)}$$

Since

the set $S = \{v_1, v_2, \dots, v_n\}$ is linearly independent

So each u_i can be uniquely expressed as a linear combination of $v_1, v_2, \dots, v_n$. Hence

$$\left. \begin{aligned} u_1 &= a_{11}v_1 + a_{12}v_2 + \dots + a_{1n}v_n \\ u_2 &= a_{21}v_1 + a_{22}v_2 + \dots + a_{2n}v_n \\ &\vdots \\ u_r &= a_{r1}v_1 + a_{r2}v_2 + \dots + a_{rn}v_n \end{aligned} \right\} \quad \text{--- (2)}$$

where $a_{ij} \in F$

Putting values of $u_1, u_2, \dots, u_r$ from (2) in (1) 57

$$C_1(a_{11}V_1 + a_{12}V_2 + \dots + a_{1n}V_n) + C_2(a_{21}V_1 + a_{22}V_2 + \dots + a_{2n}V_n) + \dots + C_k(a_{k1}V_1 + a_{k2}V_2 + \dots + a_{kn}V_n) = 0$$

2.

$$(c_1 a_{11} + c_2 a_{21} + \dots + c_\lambda a_{\lambda 1}) v_1 + (c_1 a_{12} + c_2 a_{22} + \dots + c_\lambda a_{\lambda 2}) v_2 + \dots + (c_1 a_{1n} + c_2 a_{2n} + \dots + c_\lambda a_{\lambda n}) v_n = 0$$

Since $V_1, V_2, \dots, V_n$ are linearly independent

$$\begin{aligned} a_{11}c_1 + a_{12}c_2 + \dots + a_{1n}c_n &= 0 \\ a_{21}c_1 + a_{22}c_2 + \dots + a_{2n}c_n &= 0 \\ &\vdots \\ &\vdots \\ a_{n1}c_1 + a_{n2}c_2 + \dots + a_{nn}c_n &= 0 \end{aligned}$$

Which is a homogeneous system of n eqs. in r unknowns $C_1, C_2, \dots, C_r$.

Since $n < \infty$, so this system has a non trivial soln.

Hence atleast one of $C_1, C_2, \dots, C_r$ is non zero
 & so from eq. ① set $B = \{u_1, u_2, \dots, u_r\}$ is
 linearly dependent.

Theorem Let $V_1, V_2, \dots, V_n$ be linearly independent in a vector space V over a field F . If v is any non zero vector in V . Then the set $\{V_1, V_2, \dots, V_n, v\}$ is linearly independent iff. v is not in the linear span $\langle V_1, V_2, \dots, V_n \rangle$.

Prp. Let $v \in \langle v_1, v_2, \dots, v_n \rangle$

then we prove $\{v_1, v_2, \dots, v_n, v\}$ is linearly independent.

Consider

$$a_1V_1 + a_2V_2 + \dots + a_nV_n + aV = 0 \quad \text{--- (1)}$$

where $a_1, a_2, \dots, a_n, a \in F$

Suppose that $\alpha \neq 0$ then from ①, we have

$$V = -\frac{1}{\alpha}(a_1v_1 + a_2v_2 + \dots + a_nv_n)$$

which shows that $V \in \langle v_1, v_2, \dots, v_n \rangle$

which is a Contradiction.

Hence $\alpha = 0$

Also Since $v_1, v_2, \dots, v_n$ are linearly independent

So eq. ① with $\alpha = 0$ implies $a_1, a_2, \dots, a_n = 0$

Hence $\{v_1, v_2, \dots, v_n, V\}$ is linearly independent set.

Conversely

Let $\{v_1, v_2, \dots, v_n, V\}$ be linearly independent set

then we prove $V \notin \langle v_1, v_2, \dots, v_n \rangle$

Suppose $V \in \langle v_1, v_2, \dots, v_n \rangle$

then V can be expressed as a linear combination of $v_1, v_2, \dots, v_n$ i.e.,

$$V = a_1v_1 + a_2v_2 + \dots + a_nv_n \quad \text{where } a_i \in F$$

or

$$a_1v_1 + a_2v_2 + \dots + a_nv_n + (-1)V = 0$$

which shows that $\{v_1, v_2, \dots, v_n, V\}$ is linearly dependent set.

which is a Contradiction

Hence $V \notin \langle v_1, v_2, \dots, v_n \rangle$

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Theorem Let V be an n -dimensional vector space over a field F . Then every set $S = \{v_1, v_2, \dots, v_n\}$ of n linearly independent vectors in V is a basis for V .

Proof

Let $v \in V$ be any non zero vector then the set

$\hat{S} = \{v_1, v_2, \dots, v_n, v\}$ is a linearly dependent

set. So we can find scalars $a_1, a_2, \dots, a_n, a$ not all zeros such that

$$a_1 v_1 + a_2 v_2 + \dots + a_n v_n + a v = 0 \quad \text{--- (1)}$$

Since $S = \{v_1, v_2, \dots, v_n\}$ is linearly independent

So $a_i = 0$ for $i = 1, 2, 3, \dots, n$

Hence $a \neq 0$

And so, eq. (1) can be written as

$$a v = -a_1 v_1 - a_2 v_2 - \dots - a_n v_n$$

or

$$v = \left(-\frac{a_1}{a}\right)v_1 + \left(-\frac{a_2}{a}\right)v_2 + \dots + \left(-\frac{a_n}{a}\right)v_n$$

which shows that v is a linear

combination of vectors $v_1, v_2, \dots, v_n$.

So S spans V .

Hence S is a linearly independent spanning set for V .

So $S = \{v_1, v_2, \dots, v_n\}$ is a basis for V .

Theorem (i). Any linearly independent set of vectors in a finite dimensional vector space V can be extended to a basis for V .

(ii) If W is a subspace of a finite dimensional vector space V then

$$\dim W \leq \dim V.$$

Moreover if $\dim W = \dim V$ then $W = V$

Proof:

Since V is finite dimensional so let $\dim V = n$

Let $S = \{v_1, v_2, \dots, v_r\}$ (where $r < n$) be a

linearly independent set of vectors in V .

Since $\dim V = n$, so the set S cannot span V .

There is a vector say $v_{r+1} \in V$ such that $v_{r+1} \notin \langle S \rangle$

& the set $S \cup \{v_{r+1}\}$ is linearly independent. This

process can be repeated $n-r$ times to get

a larger set $\{v_1, v_2, \dots, v_r, v_{r+1}, \dots, v_n\}$ which is linearly independent & so this will form a basis for V .

(ii) Proof:-

Since V is finite dimensional

suppose $\dim V = n$

then any set of $n+1$ or more vectors is linearly dependent.

Moreover since a basis of W consists of linearly independent vectors, so it cannot contain more than n elements

$$\text{Hence } \dim W \leq n = \dim V$$

$$\Rightarrow \dim W \leq \dim V$$

$$\text{If } \dim W = \dim V$$

Then every basis of W is also a basis for V .

$$\text{Hence } W = V$$

Theorem A vector space V is the direct sum of its subspaces U & W iff. each $v \in V$ can be uniquely written as

$$v = u + w \quad \text{for } u \in U, w \in W$$

Proof:-

Suppose that V is the direct sum of its subspaces U & W then by def.

$$(i) \quad V = U + W$$

$$(ii) \quad U \cap W = \{0\}$$

We want to show that each $v \in V$ can be uniquely written as

$$v = u + w \quad \text{for } u \in U, w \in W$$

Let $v \in V$ then

$$v = u + w \quad \text{--- (1) for } u \in U, w \in W$$

by (i)

if possible let

$$v = u_1 + w_1 \quad \text{for } u_1 \in U, w_1 \in W$$

then

$$u + w = u_1 + w_1$$

$$\text{or } u - u_1 = w_1 - w \in U \cap W$$

$$\text{But } U \cap W = \{0\}$$

by (ii)

$$\text{So } u - u_1 = w_1 - w = 0$$

$$\Rightarrow u = u_1 \text{ and } w = w_1$$

So the expression for v in (1) is unique.

Hence each $v \in V$ can be uniquely written as

$$v = u + w \quad \text{for } u \in U, w \in W$$

Conversely

Let each $v \in V$ is uniquely written as

$$v = u + w \quad \text{for } u \in U, w \in W$$

So (i) is satisfied. Now we prove Condition (ii)

For this let $v \in U \cap W$ then v can be written as

$$v = u + 0 \quad u \in U$$

$$v = 0 + w \quad w \in W$$

Since the expression for v is unique

$$\text{So } u + 0 = 0 + w$$

$$\Rightarrow u = 0, w = 0$$

$$\text{Hence } v = u + w = 0$$

$$\text{So } U \cap W = \{0\}$$

Hence Condition (ii) is satisfied.

So V is the direct sum of U & W

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Theorem If U & W are finite dimensional subspaces of a vector space V over a field F then

$$(i) \dim(U+W) = \dim U + \dim W - \dim(U \cap W)$$

$$(iii) \text{ If } U \cap W = \{0\} \text{ then } V = U \oplus W$$

$$\text{and } \dim V = \dim U + \dim W$$

Proof:

Suppose that $U \cap W \neq \{0\}$.

Let $\{v_1, v_2, \dots, v_r\}$ be a basis for $U \cap W$,

$\{v_1, v_2, \dots, v_r, u_1, u_2, \dots, u_s\}$ be a basis for U
 & $\{v_1, v_2, \dots, v_r, w_1, w_2, \dots, w_t\}$ be a basis for W
 Thus dimensions of $U \cap W, U$ & W are $r, r+s$ & $r+t$ resp.

clearly

$\{v_1, v_2, \dots, v_r, u_1, u_2, \dots, u_s, w_1, w_2, \dots, w_t\}$ spans $U+W$.

Now we show that $\{v_1, v_2, \dots, v_r, u_1, u_2, \dots, u_s, w_1, \dots, w_t\}$ is linearly independent. Suppose that

$$a_1 v_1 + a_2 v_2 + \dots + a_r v_r + b_1 u_1 + b_2 u_2 + \dots + b_s u_s + c_1 w_1 + \dots + c_t w_t = 0 \quad (1)$$

where a 's, b 's, c 's $\in F$

$$\text{or } a_1 v_1 + a_2 v_2 + \dots + a_r v_r + b_1 u_1 + b_2 u_2 + \dots + b_s u_s = -(c_1 w_1 + \dots + c_t w_t) \quad (2)$$

from eq. (2) we see that

$$-(c_1 w_1 + c_2 w_2 + \dots + c_t w_t) \in U \cap W$$

But $\{v_1, v_2, \dots, v_r\}$ is a basis for $U \cap W$. So

$$-(c_1 w_1 + c_2 w_2 + \dots + c_t w_t) = d_1 v_1 + d_2 v_2 + \dots + d_r v_r$$

where d 's $\in F$

or

$$d_1 v_1 + d_2 v_2 + \dots + d_r v_r + c_1 w_1 + c_2 w_2 + \dots + c_t w_t = 0$$

Since $\{v_1, v_2, \dots, v_r, w_1, w_2, \dots, w_t\}$ being basis of W is linearly independent

$$\Rightarrow d_1 = d_2 = \dots = d_r = 0, c_1 = c_2 = \dots = c_t = 0$$

Then eq. (1) becomes

$$a_1 v_1 + a_2 v_2 + \dots + a_r v_r + b_1 u_1 + \dots + b_s u_s = 0$$

Again $\{v_1, v_2, \dots, v_r, u_1, u_2, \dots, u_s\}$ being basis of U is linearly independent.

So $a_1 = a_2 = \dots = a_r = 0$, $b_1 = b_2 = \dots = b_s = 0$

Hence eq. (1) shows that the set

$\{v_1, v_2, \dots, v_r, u_1, u_2, \dots, u_s, w_1, w_2, \dots, w_t\}$ is linearly independent set & so it forms a basis for $U+W$. So

$$\begin{aligned} \dim(U+W) &= r+s+t \\ &= (r+s) + (r+t) - r \end{aligned}$$

$$\therefore \boxed{\dim(U+W) = \dim U + \dim W - \dim(U \cap W)}$$

(ii) Now Suppose that $U \cap W = \{0\}$

Let $\{u_1, u_2, \dots, u_s\}$ be a basis for U

& $\{w_1, w_2, \dots, w_t\}$ be a basis for W

Then clearly the set $\{u_1, u_2, \dots, u_s, w_1, w_2, \dots, w_t\}$ spans $U+W = U \oplus W$

Consider $a_1 u_1 + \dots + a_s u_s + b_1 w_1 + \dots + b_t w_t = 0$ — (2)

Then $a_1 u_1 + \dots + a_s u_s = -(b_1 w_1 + \dots + b_t w_t)$

This shows that each vector is in $U \cap W$.

Hence $a_1 = a_2 = \dots = a_s = 0$, $b_1 = b_2 = \dots = b_t = 0$

So (2) shows that $\{u_1, \dots, u_s, w_1, \dots, w_t\}$ is linearly independent & so forms a basis for $U \oplus W$

Hence $\dim(U \oplus W) = s+t$

$$\text{or } \dim(V) = \dim U + \dim W \quad \because V = U \oplus W$$

Q1 Determine whether the following vectors in $\mathbb{R}^4$ are linearly independent or linearly dependent:

(i) $(1, 3, -1, 4), (3, 8, -5, 7), (2, 9, 4, 23)$

Sol.

Let $a(1, 3, -1, 4) + b(3, 8, -5, 7) + c(2, 9, 4, 23) = 0$ $a, b, c \in F$

or $(a, 3a, -a, 4a) + (3b, 8b, -5b, 7b) + (2c, 9c, 4c, 23c) = 0$

$(a + 3b + 2c, 3a + 8b + 9c, -a - 5b + 4c, 4a + 7b + 23c) = 0$

$\Rightarrow$

$a + 3b + 2c = 0$ ——— ①

$3a + 8b + 9c = 0$ ——— ②

$-a - 5b + 4c = 0$ ——— ③

$4a + 7b + 23c = 0$ ——— ④

From ① & ③

$\frac{a}{27-16} = \frac{-b}{9-6} = \frac{c}{8-9}$

or $\frac{a}{11} = \frac{b}{-3} = \frac{c}{-1} = k$

$\Rightarrow a = 11k$

$b = -3k$

$c = -k$

Putting these values in ② & ④, we see eqs.

③ & ④ are satisfied.

Hence given vectors in $\mathbb{R}^4$ are linearly dependent.

(ii) $(1, -2, 4, 1), (2, 1, 0, -3), (1, -6, 1, 4)$

Sol.

Let $a(1, -2, 4, 1) + b(2, 1, 0, -3) + c(1, -6, 1, 4) = 0$ where $a, b, c \in F$

or $(a, -2a, 4a, a) + (2b, b, 0, -3b) + (c, -6c, c, 4c) = 0$

or $(a + 2b + c, -2a + b - 6c, 4a + c, a - 3b + 4c) = 0$

$\Rightarrow$

$$a + 2b + c = 0 \quad \text{--- (1)}$$

$$-2a + b - 6c = 0 \quad \text{--- (2)}$$

$$4a + c = 0 \quad \text{--- (3)}$$

$$a - 3b + 4c = 0 \quad \text{--- (4)}$$

From (1) & (2)

$$\frac{a}{-12-1} = \frac{-b}{-6+2} = \frac{c}{1+4}$$

$$\frac{a}{-13} = \frac{b}{4} = \frac{c}{5} = k$$

$$\Rightarrow a = -13k$$

$$b = 4k$$

$$c = 5k$$

Putting these values in (3) & (4), we see that eqs. are not satisfied. They are satisfied only when $k = 0$

$$\Rightarrow a = b = c = 0$$

Hence given vectors in $\mathbb{R}^4$ are linearly independent.

Q2 Let $V = P_3(x)$ be the vector space of all polynomials of degree ≤ 3 over $\mathbb{R}$ together with the zero polynomial. Determine whether $u, v, w \in V$ are linearly dependent or linearly independent.

$$(i) \quad u = x^3 - 4x^2 + 2x + 3, \quad v = x^3 + 2x^2 + 4x - 1, \quad w = 2x^3 - x^2 - 3x + 3$$

Sol.

$$\text{Let } au + bv + cw = 0$$

where $a, b, c \in \mathbb{F}$

$$\text{or } a(x^3 - 4x^2 + 2x + 3) + b(x^3 + 2x^2 + 4x - 1) + c(2x^3 - x^2 - 3x + 3) = 0$$

$$\text{or } (a+b+2c)x^3 + (-4a+2b-c)x^2 + (2a+4b-3c)x + (3a-b+3c) = 0$$

$$\Rightarrow a + b + 2c = 0 \quad \text{--- (1)}$$

$$-4a + 2b - c = 0 \quad \text{--- (2)}$$

$$2a + 4b - 3c = 0 \quad \text{--- (3)}$$

$$3a - b + 3c = 0 \quad \text{--- (4)}$$

From (1) & (2)

$$\frac{a}{-1-4} = \frac{-b}{-1+8} = \frac{c}{2+4}$$

$$\frac{a}{-5} = \frac{b}{-7} = \frac{c}{6} = k$$

$$\Rightarrow a = -5k$$

$$b = -7k$$

$$c = 6k$$

Putting these values in (3) & (4) we see eqs. are not satisfied. They are satisfied only when $k = 0$

$$\Rightarrow a = b = c = 0$$

Hence vectors u, v, w are linearly independent.

$$(ii) u = x^3 - 3x^2 - 2x + 3, v = x^3 - 4x^2 - 3x + 4, w = 2x^3 - 7x^2 - 7x + 9$$

Sol

$$\text{Let } au + bv + cw = 0 \quad \text{where } a, b, c \in F$$

$$\text{or } a(x^3 - 3x^2 - 2x + 3) + b(x^3 - 4x^2 - 3x + 4) + c(2x^3 - 7x^2 - 7x + 9) = 0$$

$$\text{or } (a+b+2c)x^3 + (-3a-4b-7c)x^2 + (-2a-3b-7c)x + (3a+4b+9c) = 0$$

$$\Rightarrow a+b+2c = 0 \quad \text{--- (1)}$$

$$-3a-4b-7c = 0 \quad \text{--- (2)}$$

$$-2a-3b-7c = 0 \quad \text{--- (3)}$$

$$3a+4b+9c = 0 \quad \text{--- (4)}$$

From (1) & (2)

$$\frac{a}{-7+8} = \frac{-b}{-7+6} = \frac{c}{-4+3}$$

$$\text{or } \frac{a}{1} = \frac{b}{1} = \frac{c}{-1} = k$$

$$\Rightarrow a = k$$

$$b = k$$

$$c = -k$$

Putting these values in (3) & (4), we see eqs. are not satisfied. They are satisfied only when $k = 0$

$$\Rightarrow a = b = c = 0$$

Hence given vectors u, v, w are linearly independent.

Q3 Show that the vectors $(1-i, i)$ & $(2, -1+i)$ in $\mathbb{C}^2$ are linearly dependent over $\mathbb{C}$ but linearly independent over $\mathbb{R}$.

Sol.

$$\text{Let } a(1-i, i) + b(2, -1+i) = 0$$

$$\text{or } (a(1-i), ai) + (2b, b(-1+i)) = 0$$

$$\text{or } (a(1-i) + 2b, ai + b(-1+i)) = 0$$

$$\Rightarrow a(1-i) + 2b = 0 \quad \text{--- (1)}$$

$$ai + (-1+i)b = 0 \quad \text{--- (2)}$$

These eqs. are satisfied in $\mathbb{R}$ only when $a=b=0$

Hence given vectors are linearly independent over $\mathbb{R}$.

Now we find the values of a & b from the set $\mathbb{C}$ which satisfy eqs. (1) & (2)

From (1)

$$a(1-i) = -2b$$

$$\frac{a}{b} = \frac{-2}{1-i}$$

$$= \frac{-2}{1-i} \times \frac{1+i}{1+i}$$

$$= \frac{-2(1+i)}{1+1}$$

$$= \frac{-2(1+i)}{2}$$

$$\frac{a}{b} = -(1+i)$$

$$\text{or } \frac{a}{1+i} = \frac{b}{-1} = k$$

$$\Rightarrow a = (1+i)k$$

$$b = -k$$

Putting these values in eq. (2), we see eq. (2) is satisfied.

Hence given vectors are linearly dependent over $\mathbb{C}$

Q4 Show that the vectors $(3+\sqrt{2}, 1+\sqrt{2})$ & $(7, 1+2\sqrt{2})$ in $\mathbb{R}^2$ are linearly dependent over $\mathbb{R}$ but linearly independent over $\mathbb{Q}$.

Sol.

$$\text{Let } a(3+\sqrt{2}, 1+\sqrt{2}) + b(7, 1+2\sqrt{2}) = 0$$

$$\text{or } (a(3+\sqrt{2}), a(1+\sqrt{2})) + (7b, b(1+2\sqrt{2})) = 0$$

$$\text{or } (a(3+\sqrt{2}) + 7b, a(1+\sqrt{2}) + b(1+2\sqrt{2})) = 0$$

$\Rightarrow$

$$(3+\sqrt{2})a + 7b = 0 \quad \text{--- (1)}$$

$$\& (1+\sqrt{2})a + (1+2\sqrt{2})b = 0 \quad \text{--- (2)}$$

These eqs. are satisfied in $\mathbb{Q}$ only when $a=b=0$

Hence the given vectors are linearly independent over $\mathbb{Q}$.

Now we will find values of a & b in $\mathbb{R}$ which satisfy the eqs. (1) & (2)

From (1)

$$(3+\sqrt{2})a = -7b$$

$$\text{or } \frac{a}{b} = \frac{-7}{3+\sqrt{2}}$$

$$\Rightarrow \frac{a}{7} = \frac{-b}{3+\sqrt{2}} = k$$

$$\Rightarrow a = 7k$$

$$\& b = -(3+\sqrt{2})k$$

Putting these values in eq. (2), we see eq. (2) is satisfied.

Hence given vectors are dependent over $\mathbb{R}$.

Q5 Suppose that u, v & w are linearly independent vectors. Prove that

- (i) $u+v-2w, u-v-w, u+w$ are linearly independent.
- (ii) $u+v-3w, u+3v-w, u+w$ are linearly independent.

(i) Sol.

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$$\text{Let } a(u+v-2w) + b(u-v-w) + c(u+w) = 0$$

$$\text{or } (a+b+c)u + (a-b)v + (-2a-b+c)w = 0$$

But u, v & w are linearly independent

$$\text{So } a+b+c = 0 \quad \text{--- ①}$$

$$a-b = 0 \quad \text{--- ②}$$

$$-2a-b+c = 0 \quad \text{--- ③}$$

From ① & ③

$$\frac{a}{\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 1 & 1 \\ -2 & -1 \end{vmatrix}}$$

or

$$\frac{a}{1+1} = \frac{-b}{1+2} = \frac{c}{-1+2}$$

$$\frac{a}{2} = \frac{b}{-3} = \frac{c}{1} = k$$

$$\Rightarrow a = 2k$$

$$b = -3k$$

$$c = k$$

Putting these values in ②, we see that eq.

② is not satisfied. It is satisfied only when $k = 0$

$$\Rightarrow a = b = c = 0$$

Hence given vectors are linearly independent.

(ii) Sol.

$$\text{Let } a(u+v-3w) + b(u+3v-w) + c(u+w) = 0$$

$$\text{or } (a+b+c)u + (a+3b)v + (-3a-b+c)w = 0$$

But u, v & w are linearly independent

$$\Rightarrow a+b+c = 0 \quad \text{--- ①}$$

$$a+3b = 0 \quad \text{--- ②}$$

$$-3a-b+c = 0 \quad \text{--- ③}$$

From ① & ③

$$\frac{a}{\begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix}} = \frac{-b}{\begin{vmatrix} 1 & 1 \\ -3 & 1 \end{vmatrix}} = \frac{c}{\begin{vmatrix} 1 & 1 \\ -3 & -1 \end{vmatrix}}$$

$$\frac{a}{1+1} = \frac{-b}{1+3} = \frac{c}{-1+3}$$

$$\frac{a}{2} = \frac{b}{-4} = \frac{c}{2} = K$$

$$\Rightarrow a = 2K$$

$$b = -4K$$

$$c = 2K$$

Putting these values in eq. ②, we see eq. ② is not satisfied. It is satisfied only if $K=0$

$$\Rightarrow a = b = c = 0$$

Hence given vectors are linearly independent.

Q1. Determine K so that the vectors $(1, -1, K-1), (2, K, -4), (0, 2+K, -8)$ in $\mathbb{R}^3$ are linearly dependent.

Sol. Suppose that given vectors are linearly dependent then one of them must be a linear combination of the other two

$$\text{Let } (1, -1, K-1) = a(2, K, -4) + b(0, 2+K, -8) \text{ where } a, b \in F$$

$$= (2a, aK, -4a) + (0, b(2+K), -8b)$$

$$\text{or } (1, -1, K-1) = (2a, aK + b(2+K), -4a - 8b)$$

$$\Rightarrow 2a = 1 \quad \text{--- ①}$$

$$aK + b(2+K) = -1 \quad \text{--- ②}$$

$$-4a - 8b = K-1 \quad \text{--- ③}$$

$$\text{from ① } a = \frac{1}{2}$$

Put in ② & ③

$$\text{②} \Rightarrow \frac{K}{2} + b(2+K) = -1 \quad \text{--- ④}$$

$$\text{③} \Rightarrow -4\left(\frac{1}{2}\right) - 8b = K-1$$

$$-2 - 8b = K-1$$

$$-8b = K - 1 + 2$$

$$-8b = K + 1$$

$$b = -\frac{K+1}{8}$$

Put this value in (4)

$$\frac{K}{2} + (2+K)\left(-\frac{K+1}{8}\right) = -1$$

$$4K - (2+K)(K+1) = -8$$

$$4K - 2K - 2 - K^2 - K + 8 = 0$$

$$-K^2 + K + 6 = 0$$

$$\text{or } K^2 - K - 6 = 0$$

$$K^2 - 3K + 2K - 6 = 0$$

$$K(K-3) + 2(K-3) = 0$$

$$(K-3)(K+2) = 0$$

$$\Rightarrow \boxed{K = 3, -2}$$

So for $K = 3, -2$; given vectors are linearly dependent.

Q1 Using the technique of casting out vectors which are linear combination of others, find a linearly independent subset of the given set spanning the same subspace:

$$(i) \{(1, -3, 1), (2, 1, -4), (-2, 6, -2), (-1, 10, -7)\} \text{ in } \mathbb{R}^3$$

Sol.

$$\text{Given set is } \{(1, -3, 1), (2, 1, -4), (-2, 6, -2), (-1, 10, -7)\}$$

We see that

$$(-2, 6, -2) = -2(1, -3, 1)$$

So we cast out $(-2, 6, -2)$ & obtain the subset

$$\{(1, -3, 1), (2, 1, -4), (-1, 10, -7)\}$$

$$\text{Suppose } (-1, 10, -7) = a(1, -3, 1) + b(2, 1, -4) \quad \text{for } a, b \in \mathbb{R}$$

$$\text{or } (-1, 10, -7) = (a+2b, -3a+b, a-4b)$$

$$\Rightarrow \begin{aligned} a+2b &= -1 & \text{--- (1)} \\ -3a+b &= 10 & \text{--- (2)} \\ a-4b &= -7 & \text{--- (3)} \end{aligned}$$

Multiplying ① by 2 & adding in ③

$$2a + 4b = -2 \quad \text{--- ④}$$

$$a - 4b = -7 \quad \text{--- ③}$$

$$3a = -9$$

$$a = -3$$

Put in ③

$$-3 - 4b = -7$$

$$-4b = -7 + 3$$

$$-4b = -4$$

$$b = 1$$

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$$\text{So } (-1, 10, -7) = -3(1, -3, 1) + 1(2, 1, -4)$$

Hence $(-1, 10, -7)$ is a linear combination of $(1, -3, 1)$ & $(2, 1, -4)$

We cast out $(-1, 10, -7)$ & obtain a subset

$$A = \{(1, -3, 1), (2, 1, -4)\}$$

Since $(2, 1, -4)$ is not a multiple of $(1, -3, 1)$ so

the set $A = \{(1, -3, 1), (2, 1, -4)\}$ is a req. linearly independent set which spans the same subspace as the given set of four vectors.

(ii) $\{1, \sin^2 x, \cos 2x, \cos^2 x\}$ in the space of all functions from $\mathbb{R}$ to $\mathbb{R}$.

Sol.

Given set is $\{1, \sin^2 x, \cos 2x, \cos^2 x\}$

As

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\text{or } \cos 2x = 1 \cdot \cos^2 x + (-1) \cdot \sin^2 x$$

So $\cos 2x$ is a linear combination of $\cos^2 x$ & $\sin^2 x$

We cast out $\cos 2x$ & obtain the subset

$$\{1, \sin^2 x, \cos^2 x\}$$

$$\text{Also } \cos^2 x = 1 - \sin^2 x$$

$$\text{or } \cos^2 x = 1 \cdot 1 + (-1) \sin^2 x$$

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So $\cos^2 x$ is a linear combination of 1 & $\sin^2 x$

We cast out $\cos^2 x$ & obtain the subset $\{1, \sin^2 x\}$

Since none of 1 & $\sin^2 x$ is multiple of other

So $\{1, \sin^2 x\}$ is req. linearly independent set which spans the same subspace as the given set of four vectors.

(iii) $\{1, 3x-4, 4x+3, x^2+2, x-x^2\}$ in the space $P_2(x)$ of polynomials.

Sol.

Given set is $\{1, 3x-4, 4x+3, x^2+2, x-x^2\}$

We see that

$$3x-4 = \frac{3}{4}(4x+3) - \frac{25}{4}(1)$$

So $3x-4$ is a linear combination of $4x+3$ & 1

We cast out $3x-4$ & obtain the subset $\{1, 4x+3, x^2+2, x-x^2\}$

Now

$$\text{Let } x-x^2 = a(1) + b(4x+3) + c(x^2+2)$$

$$\text{or } x-x^2 = (a+3b+2c) + (4b)x + cx^2$$

$$\Rightarrow a+3b+2c = 0 \quad \text{--- (1)}$$

$$4b = 1 \quad \text{--- (2)}$$

$$c = -1 \quad \text{--- (3)}$$

$$\textcircled{2} \Rightarrow \boxed{c = -1}$$

$$\textcircled{2} \Rightarrow \boxed{b = \frac{1}{4}}$$

$$\textcircled{1} \Rightarrow a + \frac{3}{4} - 2 = 0$$

$$a - \frac{5}{4} = 0$$

$$\boxed{a = \frac{5}{4}}$$

$$\text{So } x-x^2 = \frac{5}{4}(1) + \frac{1}{4}(4x+3) - 1(x^2+2)$$

Hence $x - x^2$ is a linear combination of $1, 4x+3$ & x^2+2

We cast out $x - x^2$ & obtain the subset

$$A = \{1, 4x+3, x^2+2\}$$

We check whether A is linearly independent or not

$$\text{Let } a(1) + b(4x+3) + c(x^2+2) = 0$$

$$\text{or } (a+3b+2c) + 4bx + cx^2 = 0$$

$$\Rightarrow a + 3b + 2c = 0 \quad \text{--- (1)}$$

$$4b = 0 \quad \text{--- (2)}$$

$$c = 0 \quad \text{--- (3)}$$

$$\text{(3)} \Rightarrow c = 0$$

$$\text{(2)} \Rightarrow b = 0$$

Put in (1)

$$a + 0 + 0 = 0 \Rightarrow a = 0$$

Hence the set $A = \{1, 4x+3, x^2+2\}$ is linearly independent set which spans the same subspace as the given set of five vectors.

Q3 Verify that the polynomials $2-x^2, x^3-x, 2-3x^2$ & $3-x^3$ form a basis for $P_3(x)$. Express each of (i) $1+x$ & (ii) $x+x^2$ as a linear combination of these basis vectors.

Sol.

We want to show that $\{2-x^2, x^3-x, 2-3x^2, 3-x^3\}$ form a basis for $P_3(x)$.

First we prove that this set is linearly independent.

$$\text{Let } a(2-x^2) + b(x^3-x) + c(2-3x^2) + d(3-x^3) = 0 \quad a, b, c, d \in F$$

$$\text{or } (2a+2c+3d) - bx + (-a-3c)x^2 + (b-d)x^3 = 0$$

$$\Rightarrow 2a + 2c + 3d = 0 \quad \text{--- (1)}$$

$$-b = 0 \quad \text{--- (2)}$$

$$-a - 3c = 0 \quad \text{--- (3)}$$

$$b - d = 0 \quad \text{--- (4)}$$

$$\textcircled{2} \Rightarrow \boxed{b=0}$$

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$$\textcircled{3} \Rightarrow \boxed{b=d}$$

$$\textcircled{4} \Rightarrow a = -3c$$

Put in $\textcircled{1}$

$$2(-3c) + 2c + 3(0) = 0$$

$$-6c + 2c = 0$$

$$-4c = 0 \Rightarrow \boxed{c=0}$$

Put in $\textcircled{2}$

$$-a - 3(0) = 0 \Rightarrow \boxed{a=0}$$

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Hence the given set of polynomials is linearly independent

As dimension of $P_3(x)$ is 4

& no. of vectors in set $\{2-x^2, x^3-x, 2-3x^2, 3-x^3\}$

So the given set $\{2-x^2, x^3-x, 2-3x^2, 3-x^3\}$ form a basis for $P_3(x)$.

Now we express $1+x$ as a linear combination of given vectors.

$$\text{Let } 1+x = a(2-x^2) + b(x^3-x) + c(2-3x^2) + d(3-x^3)$$

$$\text{or } 1+x = (2a+2c+3d) - bx + (-a-3c)x^2 + (b-d)x^3$$

where $a, b, c, d \in \mathbb{R}$

$$\Rightarrow 2a + 2c + 3d = 1 \quad \text{---} \textcircled{1}$$

$$-b = 1 \quad \text{---} \textcircled{2}$$

$$-a - 3c = 0 \quad \text{---} \textcircled{3}$$

$$b - d = 0 \quad \text{---} \textcircled{4}$$

$$\textcircled{2} \Rightarrow \boxed{b=-1}$$

Put in $\textcircled{4}$

$$-1 - d = 0 \Rightarrow \boxed{d=-1}$$

$$\textcircled{3} \Rightarrow a = -3c$$

Put in $\textcircled{1}$

$$2(-3c) + 2c + 3(-1) = 1$$

$$-4c - 3 = 1$$

$$-4c = 4$$

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$$\Rightarrow \boxed{c = -1}$$

Put in (3)

$$-a - 3(-1) = 0$$

$$-a + 3 = 0 \Rightarrow \boxed{a = 3}$$

$$\text{So } 1+x = 3(2-x^2) - (x^3-x) - (2-3x^2) - (3-x^3)$$

(ii) Now we express $x+x^2$ as a linear combination of given vectors

$$\text{Let } x+x^2 = a(2-x^2) + b(x^3-x) + c(2-3x^2) + d(3-x^3)$$

where $a, b, c, d \in \mathbb{R}$

$$\text{or } x+x^2 = (2a+2c+3d) - bx + (-a-3c)x^2 + (b-d)x^3$$

$$\Rightarrow 2a+2c+3d = 0 \quad \text{--- (1)}$$

$$-b = 1 \quad \text{--- (2)}$$

$$-a-3c = 1 \quad \text{--- (3)}$$

$$b-d = 0 \quad \text{--- (4)}$$

$$\text{(2)} \Rightarrow \boxed{b = -1}$$

$$\text{(4)} \Rightarrow -1-d = 0 \text{ or } \boxed{d = -1}$$

$$\text{(3)} \Rightarrow a = -1-3c$$

Put in (1)

$$2(-1-3c) + 2c + 3(-1) = 0$$

$$-2 - 6c + 2c - 3 = 0$$

$$-4c - 5 = 0$$

$$4c = -5 \Rightarrow \boxed{c = -\frac{5}{4}}$$

Put in (3)

$$-a - 3(-\frac{5}{4}) = 1$$

$$-a + \frac{15}{4} = 1$$

$$a = \frac{15}{4} - 1$$

$$\boxed{a = \frac{11}{4}}$$

$$\text{So } x+x^2 = \frac{11}{4}(2-x^2) - (x^3-x) - \frac{5}{4}(2-3x^2) - (3-x^3)$$

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Q9 Determine whether or not the given set of 77
vectors is a basis for $\mathbb{R}^2$.

(i) $\{(1,1), (3,1)\}$

Sol. Given set is $\{(1,1), (3,1)\}$

First we will check their independency
For scalars $a, b \in \mathbb{R}$

Let $a(1,1) + b(3,1) = 0$

or $(a+3b, a+b) = 0$

$\Rightarrow$

$a+3b = 0$ ----- ①

$a+b = 0$ ----- ②

Sol. ② from ①

$2b = 0 \Rightarrow \boxed{b=0}$

Put in ①

$a+0 = 0 \Rightarrow \boxed{a=0}$

Hence given set of vectors is linearly independent.

Since dimension of $\mathbb{R}^2$ is 2

& the linearly independent vectors are also 2.

So the given set of vectors $\{(1,1), (3,1)\}$ form a basis for $\mathbb{R}^2$

(ii) $\{(2,1), (1,-1)\}$

Sol. Given set is $\{(2,1), (1,-1)\}$

First we will check their independency.

For scalars $a, b \in \mathbb{R}$

Let $a(2,1) + b(1,-1) = 0$

or $(2a+b, a-b) = 0$

$\Rightarrow 2a+b = 0$ ----- ①

& $a-b = 0$ ----- ②

Adding ① & ②

$3a = 0 \Rightarrow \boxed{a=0}$

$$0 - b = 0 \Rightarrow \boxed{b = 0}$$

Hence given set of vectors $\{(2,1), (1,-1)\}$ is linearly independent.

Since dimension of $\mathbb{R}^2$ is 2

& no. of linearly independent vectors are also 2

So $\{(2,1), (1,-1)\}$ forms a basis for $\mathbb{R}^2$

Q10 Determine whether or not the given set of vectors is a basis for $\mathbb{R}^3$:

(i) $\{(1,2,-1), (0,3,1), (1,-5,3)\}$

Sol. Given set is $\{(1,2,-1), (0,3,1), (1,-5,3)\}$

First we will check their independency.

For scalars $a, b, c \in \mathbb{R}$

$$\text{Let } a(1,2,-1) + b(0,3,1) + c(1,-5,3) = 0$$

$$\text{or } (a+c, 2a+3b-5c, -a+b+3c) = 0$$

$$\Rightarrow \begin{aligned} a+c &= 0 & \text{--- ①} \\ 2a+3b-5c &= 0 & \text{--- ②} \\ -a+b+3c &= 0 & \text{--- ③} \end{aligned}$$

From ② & ③

$$\frac{a}{9+5} = \frac{-b}{6-5} = \frac{c}{2+3}$$

$$\frac{a}{14} = \frac{b}{-1} = \frac{c}{5} = k$$

$$\Rightarrow a = 14k$$

$$b = -k$$

$$c = 5k$$

Putting these values in ①, we see eq. ① is not satisfied. It is satisfied only when $k = 0$

$$\Rightarrow a = 0, b = 0, c = 0$$

Hence given set of vectors $\{(1,2,-1), (0,3,1), (1,-5,3)\}$ is linearly independent.

Since dimension of R^3 is 3

& the no. of linearly independent vectors in R^3 is also 3.

So the set $\{(1, 2, -1), (0, 3, 1), (1, -5, 2)\}$ forms a basis for R^3

(ii) $\{(2, 4, -3), (0, 1, 1), (0, 1, -1)\}$

Soln

Given set is $\{(2, 4, -3), (0, 1, 1), (0, 1, -1)\}$

First we will check their independency.

For scalars $a, b, c \in R$

Let $a(2, 4, -3) + b(0, 1, 1) + c(0, 1, -1) = 0$

$$\Rightarrow (2a, 4a + b + c, -3a + b - c) = 0$$

$$\Rightarrow \begin{aligned} 2a &= 0 & \text{--- (1)} \end{aligned}$$

$$4a + b + c = 0 & \text{--- (2)}$$

$$-3a + b - c = 0 & \text{--- (3)}$$

$$\textcircled{1} \Rightarrow \boxed{a = 0}$$

Put in (2) & (3)

$$b + c = 0 & \text{--- (4)}$$

$$b - c = 0 & \text{--- (5)}$$

Add (4) & (5)

$$2b = 0 \Rightarrow \boxed{b = 0}$$

Put in (2)

$$0 + c = 0 \Rightarrow \boxed{c = 0}$$

Hence given set of vectors $\{(2, 4, -3), (0, 1, 1), (0, 1, -1)\}$ is linearly independent.

Since dimension of R^3 is 3

& no. of linearly independent vectors in R^3 is also 3

So the set $\{(2, 4, -3), (0, 1, 1), (0, 1, -1)\}$ forms a basis for R^3

Q11 Let V be the real vector space of all functions defined on $\mathbb{R}$ into $\mathbb{R}$. Determine whether the given vectors are linearly independent or linearly dependent in V :

(i) $x, \cos x$

Sol. Given vectors are $x, \cos x$

Suppose that for scalars $a, b \in \mathbb{R}$

$$ax + b\cos x = 0 \quad \text{--- (1) for all } x \in \mathbb{R}$$

Put $x = 0$ in (1)

$$a(0) + b\cos 0 = 0$$

$$0 + b = 0 \Rightarrow \boxed{b = 0}$$

Now Put $x = \pi/2$ in (1)

$$a(\pi/2) + b\cos \pi/2 = 0$$

$$a(\pi/2) + 0 = 0$$

$$\boxed{a = 0}$$

Hence given vectors are linearly independent.

(ii) $\sin^2 x, \cos^2 x, \cos 2x$

Sol. Given vectors are $\sin^2 x, \cos^2 x, \cos 2x$

Suppose that for scalars $a, b, c \in \mathbb{R}$

$$a\sin^2 x + b\cos^2 x + c\cos 2x = 0 \quad \text{--- (1) for all } x \in \mathbb{R}$$

Put $x = 0$ in (1)

$$a\sin^2(0) + b\cos^2 0 + c\cos 0 = 0$$

$$b + c = 0 \quad \text{--- (2)}$$

Put $x = \pi/2$ in (1)

$$a\sin^2 \frac{\pi}{2} + b\cos^2 \frac{\pi}{2} + c\cos \pi = 0$$

$$a - c = 0 \quad \text{--- (3)}$$

Put $x = \pi/4$ in (1)

$$a\sin^2 \pi/4 + b\cos^2 \pi/4 + c\cos \pi/2 = 0$$

$$a \cdot \frac{1}{2} + \frac{b}{2} = 0$$

$$\text{or } a + b = 0 \quad \text{--- (4)}$$

$$\textcircled{2} \Rightarrow b = -c$$

$$\textcircled{3} \Rightarrow a = c$$

So, non zero soln. of above eq. is

$$a = c$$

$$b = -c$$

$$c = c \quad \text{or } (c, -c, c) \quad \text{where } c \in \mathbb{R}$$

So, given vectors are linearly dependent.

(iii) $\sin x, \cos x, \sinh x, \cosh x$

Sol.

Given vectors are $\sin x, \cos x, \sinh x, \cosh x$

Suppose that for scalars $a, b, c, d \in \mathbb{R}$

$$a \sin x + b \cos x + c \sinh x + d \cosh x = 0 \quad \text{--- (A)}$$

Put $x = 0$

$$a \sin 0 + b \cos 0 + c \sinh 0 + d \cosh 0 = 0$$

$$b + d = 0 \quad \text{--- (1)}$$

Diff. (A) w.r.t. x

$$a \cos x - b \sin x + c \cosh x + d \sinh x = 0 \quad \text{--- (B)}$$

Put $x = 0$

$$a \cos 0 - b \sin 0 + c \cosh 0 + d \sinh 0 = 0$$

$$a + c = 0 \quad \text{--- (2)}$$

Diff. (B) w.r.t. x

$$-a \sin x - b \cos x + c \sinh x + d \cosh x = 0 \quad \text{--- (C)}$$

Put $x = 0$

$$-a \sin 0 - b \cos 0 + c \sinh 0 + d \cosh 0 = 0$$

$$\text{or } -b + d = 0 \quad \text{--- (2)}$$

Diff. (C) w.r.t. x

$$-a \cos x + b \sin x + c \cosh x + d \sinh x = 0$$

Put $x = 0$

$$-a \cos 0 + b \sin 0 + c \cosh 0 + d \sinh 0 = 0$$

$$-a + c = 0 \quad \text{--- (4)}$$

Adding (1) + (3)

$$2d = 0 \Rightarrow \boxed{d = 0}$$

Put in (1)

$$b + 0 = 0 \Rightarrow \boxed{b = 0}$$

Adding (2) + (4)

$$2c = 0 \Rightarrow \boxed{c = 0}$$

Put in (4)

$$-a + 0 = 0 \Rightarrow \boxed{a = 0}$$

Hence given vectors are linearly independent.

(iv) $\sin x, \sin x + \cos x, \sin x - \cos x$

Soln. Given vectors are $\sin x, \sin x + \cos x, \sin x - \cos x$

Suppose that for scalars $a, b, c \in \mathbb{R}$

$$a \sin x + b(\sin x + \cos x) + c(\sin x - \cos x) = 0$$

$$\Rightarrow (a+b+c) \sin x + (b-c) \cos x = 0 \quad \text{--- (A)}$$

Put $x = 0$ in (A)

$$(a+b+c) \sin 0 + (b-c) \cos 0 = 0$$

$$\Rightarrow b - c = 0 \quad \text{--- (1)}$$

Put $x = \pi/2$ in (A)

$$(a+b+c) \sin \pi/2 + (b-c) \cos \pi/2 = 0$$

$$\Rightarrow a + b + c = 0 \quad \text{--- (2)}$$

Put $x = \pi$ in (A)

$$(a+b+c) \sin \pi + (b-c) \cos \pi = 0$$

$$\Rightarrow (b-c)(-1) = 0$$

$$\Rightarrow b - c = 0 \quad \text{--- (3)}$$

$$\textcircled{3} \Rightarrow b = c$$

Put in (2)

$$a + c + c = 0 \Rightarrow \boxed{a = -2c}$$

So a non zero soln. of hom. eqn. is $(-2c, c, c)$

Hence the given vectors are linearly dependent. 83

(v) e^{ax}, e^{bx}, e^{cx} ; a, b, c being distinct real no's.

Sol.

Given vectors are e^{ax}, e^{bx}, e^{cx}

Suppose that for scalars $\alpha, \beta, \gamma \in \mathbb{R}$

$$\alpha e^{ax} + \beta e^{bx} + \gamma e^{cx} = 0$$



Q12. Determine a basis for each of the following subspace of $\mathbb{R}^3$:

(i) The plane $x - 2y + 5z = 0$

Sol.

Given eq. of plane is $x - 2y + 5z = 0$

or $x = 2y - 5z$ where y, z are free variables

The above eq. in vector form can be written as

$$\begin{aligned}(x, y, z) &= (2y - 5z, y, z) \\ &= (2y - 5z, y + 0, 0 + z) \\ &= (2y, y, 0) + (-5z, 0, z)\end{aligned}$$

$$\Rightarrow (x, y, z) = y(2, 1, 0) + z(-5, 0, 1)$$

Thus given plane is spanned by vectors $(2, 1, 0)$ &

$(-5, 0, 1)$. Since none of the vector is multiple of other. So the set $\{(2, 1, 0), (-5, 0, 1)\}$ is linearly independent.

Hence $\{(2, 1, 0), (-5, 0, 1)\}$ forms a basis for given subspace of $\mathbb{R}^3$.

(ii) The line $\frac{x}{-2} = \frac{y}{1} = \frac{z}{6}$

Sol.

Given eq. of line is

$$\frac{x}{-2} = \frac{y}{1} = \frac{z}{6} = t$$

$$\Rightarrow x = -2t$$

$$y = t$$

$$z = 6t$$

The above eq. in vector form can be written as

$$(x, y, z) = (-2t, t, 6t)$$

$$\text{or } (x, y, z) = t(-2, 1, 6)$$

Hence the given line is spanned by the vector $(-2, 1, 6)$. Also $(-2, 1, 6)$ being a non

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zero single vector is linearly independent 85
 So $\{(-2, 1, 6)\}$ forms a basis for the given subspace of $\mathbb{R}^3$.

(iii) All vectors of the form (a, b, c) where
 $3a - 2b + c = 0$

Sol.

Given eq. is

$$3a - 2b + c = 0$$

or $c = -3a + 2b$ where a, b are free variables

The above eq. can be written in vector form as

$$\begin{aligned}(a, b, c) &= (a, b, -3a + 2b) \\ &= (a + 0, 0 + b, -3a + 2b) \\ &= (a, 0, -3a) + (0, b, 2b)\end{aligned}$$

$$\text{or } (a, b, c) = a(1, 0, -3) + b(0, 1, 2)$$

So given subspace is spanned by $(1, 0, -3)$ & $(0, 1, 2)$.

Now since none of the vector is multiple of other. So the vectors $(1, 0, -3)$ & $(0, 1, 2)$ are linearly independent

So the set $\{(1, 0, -3), (0, 1, 2)\}$ forms a basis for the given subspace of $\mathbb{R}^3$.

Q13 Find the dimension of the subspace

$\{(x_1, x_2, x_3, x_4) : x_2 = x_3\}$ of $\mathbb{R}^4$. Also determine a basis.

Sol.

$$\text{Let } W = \{(x_1, x_2, x_3, x_4) : x_2 = x_3\}$$

Suppose, (x_1, x_2, x_2, x_4) be a general vector of W

then we can write it as

$$\begin{aligned}(x_1, x_2, x_2, x_4) &= (x_1, 0, 0, 0) + (0, x_2, x_2, 0) + (0, 0, 0, x_4) \\ &= x_1(1, 0, 0, 0) + x_2(0, 1, 1, 0) + x_4(0, 0, 0, 1)\end{aligned}$$

which shows that $(x_1, x_2, x_3, x_4) \in W$ is a linear combination of vectors $(1, 0, 0, 0)$, $(0, 1, 1, 0)$ & $(0, 0, 0, 1)$.

So the set $S = \{(1, 0, 0, 0), (0, 1, 1, 0), (0, 0, 0, 1)\}$ spans W .

Now we check the independence of set S .

Suppose that for scalars $a, b \in \mathbb{R}$

$$a(1, 0, 0, 0) + b(0, 1, 1, 0) + c(0, 0, 0, 1) = (0, 0, 0, 0)$$

$$\text{or } (a, b, b, c) = (0, 0, 0, 0)$$

$$\Rightarrow a = 0$$

$$b = 0$$

$$c = 0$$

Hence the set $S = \{(1, 0, 0, 0), (0, 1, 1, 0), (0, 0, 0, 1)\}$ is also linearly independent.

Hence S is a basis for W .

So dimension of W is 3.

Q14 A subspace U of $\mathbb{R}^4$ is spanned by the vectors $(1, 0, 2, 3)$ & $(0, 1, -1, 2)$ & a subspace W is spanned by $(1, 2, 3, 4)$, $(-1, -1, 5, 0)$ & $(0, 0, 0, 1)$. Find the dimensions of U & W .

Sol.

$$\text{Let } S = \{(1, 0, 2, 3), (0, 1, -1, 2)\}$$

Since these vectors span U

So we only check their independence.

Suppose that for scalars $a, b \in \mathbb{R}$

$$a(1, 0, 2, 3) + b(0, 1, -1, 2) = 0$$

$$\text{or } (a, b, 2a-b, 3a+2b) = (0, 0, 0, 0)$$

$$\Rightarrow a = 0$$

$$b = 0$$

$$2a - b = 0$$

$$3a + 2b = 0$$

$$\Rightarrow a = b = 0$$

which shows that set S is linearly independent

Hence $S = \{(1, 0, 2, 3), (0, 1, -1, 2)\}$ ⁵⁹² forms a basis for U 87
 So dimension of $U = 2$

(ii) $(1, 2, 3, 4), (-1, -1, 5, 0), (0, 0, 0, 1)$

Sol.

Let $S = \{(1, 2, 3, 4), (-1, -1, 5, 0), (0, 0, 0, 1)\}$

Since these vectors span W (given)

So we only check their independency.

Suppose that for scalars $a, b \in \mathbb{R}$

$$a(1, 2, 3, 4) + b(-1, -1, 5, 0) + c(0, 0, 0, 1) = 0$$

$$\Rightarrow (a-b, 2a-b, 3a+5b, 4a+c) = (0, 0, 0, 0)$$

$\Rightarrow$

$$a - b = 0 \quad \text{--- ①}$$

$$2a - b = 0 \quad \text{--- ②}$$

$$3a + 5b = 0 \quad \text{--- ③}$$

$$4a + c = 0 \quad \text{--- ④}$$

Solnt. ② from ①

$$-a = 0 \Rightarrow a = 0$$

$$\text{①} \Rightarrow b = 0$$

$$\text{④} \Rightarrow c = 0$$

Hence the set $S = \{(1, 2, 3, 4), (-1, -1, 5, 0), (0, 0, 0, 1)\}$ is linearly independent.

So $S = \{(1, 2, 3, 4), (-1, -1, 5, 0), (0, 0, 0, 1)\}$ forms a basis for W
 Hence dimension of $W = 3$

Q15. Suppose that U & W are distinct four dimensional subspaces of a vector space V of dimension six. Find the possible dimension of $U \cap W$.

Sol. We are given that

$$\dim U = 4$$

$$\dim W = 4$$

$$\& \dim V = 6$$

Since $U+W$ is a subspace of V

$$\text{So } \dim(U+W) \leq \dim V = 6$$

$$\Rightarrow \dim(U+W) \leq 6$$

Now as $U \subseteq U+W$ & $W \subseteq U+W$

$$\text{So } \dim U \leq \dim(U+W) \leq 6$$

$$\text{or } 4 \leq \dim(U+W) \leq 6$$

Hence $\dim(U+W)$ is 4 or 5 or 6

Since U & W are distinct, so they must be different by atleast one generator

$$\text{So } \dim(U+W) > 4$$

Hence $\dim(U+W)$ is 5 or 6

As we know

$$\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$$

$$\text{or } \dim(U \cap W) = \dim U + \dim W - \dim(U+W)$$

$$(i) \text{ If } \dim(U+W) = 5$$

$$\text{Then } \dim(U \cap W) = 4 + 4 - 5 = 3$$

$$(ii) \text{ If } \dim(U+W) = 6$$

$$\text{Then } \dim(U \cap W) = 4 + 4 - 6 = 2$$

Hence the possible dimensions of $U \cap W$ are 2 or 3

Q16 Find a basis & dimension of the subspace W of $\mathbb{R}^4$ spanned by

$$(i) (1, 4, -1, 3), (2, 1, -3, -1) \text{ & } (0, 2, 1, -5)$$

$$(ii) (1, -4, -2, 1), (1, -3, -1, 2) \text{ & } (3, -8, -2, 7)$$

Sol.

As the subspace W of $\mathbb{R}^4$ is spanned by $(1, 4, -1, 3)$, $(2, 1, -3, -1)$ & $(0, 2, 1, -5)$

Now we only check their independency

For this let for $a, b, c \in F$

$$a(1, 4, -1, 3) + b(2, 1, -3, -1) + c(0, 2, 1, -5) = (0, 0, 0, 0)$$

$$(a+2b, 4a+b+2c, -a-3b+c, 3a-b-5c) = (0, 0, 0, 0)$$

$$\Rightarrow a+2b = 0 \quad \text{--- (1)}$$

$$4a+b+2c = 0 \quad \text{--- (2)}$$

$$-a-3b+c = 0 \quad \text{--- (3)}$$

$$3a-b-5c = 0 \quad \text{--- (4)}$$

from (1) & (2)

$$\frac{a}{4-0} = \frac{-b}{2-0} = \frac{c}{1-8}$$

$$\frac{a}{4} = \frac{b}{-2} = \frac{c}{-7} = k$$

$$\Rightarrow \left. \begin{aligned} a &= 4k \\ b &= -2k \\ c &= -7k \end{aligned} \right\}$$

Put in (3) & (4), we see eqs. are not satisfied & they are satisfied only when $k = 0$

i.e., when $a = b = c = 0$

Which shows that given vectors are linearly independent

S. $\{(1, 4, -1, 3), (2, 1, -3, -1), (0, 2, 1, -5)\}$ form a basis of W

$$\text{Hence } \dim W = 3$$

(ii) $(1, -4, -2, 1), (1, -3, -1, 2), (3, -8, -2, 7)$

Sol. Given vectors are

$$(1, -4, -2, 1), (1, -3, -1, 2) \text{ & } (3, -8, -2, 7)$$

As the subspace W of $\mathbb{R}^4$ is spanned by $(1, -4, -2, 1)$, $(1, -3, -1, 2)$ & $(3, -8, -2, 7)$.

So we only check their independency.

For this let for scalars a, b, c

$$a(1, -4, -2, 1) + b(1, -3, -1, 2) + c(3, -8, -2, 7) = (0, 0, 0, 0)$$

$$\text{or } (a+b+3c, -4a-3b-8c, -2a-b-2c, a+2b+7c) = (0, 0, 0, 0)$$

$$\Rightarrow a+b+3c = 0 \quad \text{--- (1)}$$

$$-4a-3b-8c = 0 \quad \text{--- (2)}$$

$$-2a-b-2c = 0 \quad \text{--- (3)}$$

$$a+2b+7c = 0 \quad \text{--- (4)}$$

from (1) & (2)

$$\frac{a}{-8+9} = \frac{-b}{-8+12} = \frac{c}{-3+4}$$

$$\frac{a}{1} = \frac{b}{-4} = \frac{c}{1} = k$$

$$\Rightarrow a = k$$

$$b = -4k$$

$$c = k$$

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Put these values in (3) & (4), we see these eqs. are satisfied so the given vectors are linearly dependent.

Now we take only first two vectors $(1, -4, -2, 1)$ & $(1, -3, -1, 2)$ & check their independency.

Suppose for $a, b \in F$

$$a(1, -4, -2, 1) + b(1, -3, -1, 2) = (0, 0, 0, 0)$$

$$\text{or } (a+b, -4a-3b, -2a-b, a+2b) = (0, 0, 0, 0)$$

$$\Rightarrow a+b = 0 \quad \text{--- (1)}$$

$$-4a-3b = 0 \quad \text{--- (2)}$$

$$-2a-b = 0 \quad \text{--- (3)}$$

$$a+2b = 0 \quad \text{--- (4)}$$

Adding ① + ③

$$-a = 0 \quad \text{or} \quad \boxed{a = 0}$$

Put in ①

$$0 + b = 0 \quad \Rightarrow \quad \boxed{b = 0}$$

Hence the vectors are linearly independent & so the set $\{(1, -4, -2, 1), (1, -3, -1, 2)\}$ form a basis of W

$$\text{Hence } \dim W = 2$$

Q17 Let U & W be 2-dimensional subspaces of $\mathbb{R}^3$. Show that $U \cap W \neq \{0\}$

Sol. Given that

$$\left. \begin{array}{l} \dim U = 2 \\ \& \dim W = 2 \end{array} \right\}$$

If $U = W$ then $\dim U = \dim W = 2$

$$\text{So } U \cap W \neq \{0\}$$

Hence we suppose that $U \neq W$. This means that U & W are not spanned by the same set.

$$\text{So } \dim(U+W) > 2$$

Since U & W are subspaces of $\mathbb{R}^3$

$$\text{Hence } \dim(U+W) \leq 3$$

$$\text{So } 2 < \dim(U+W) \leq 3$$

$$\text{Hence } \dim(U+W) = 3$$

$$\text{As } \dim(U+W) = \dim U + \dim W - \dim(U \cap W)$$

$$\begin{aligned} \text{So } \dim(U \cap W) &= \dim U + \dim W - \dim(U+W) \\ &= 2 + 2 - 3 \end{aligned}$$

$$\therefore \dim(U \cap W) = 1$$

It shows that $U \cap W$ contains a non zero element & so $U \cap W \neq \{0\}$